# RIVER VALLEY HIGH SCHOOL
## YEAR 6 PRELIMINARY EXAMINATION

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|---|---|---|---|---|---|---|---|---|---|---|
| CANDIDATE NAME | | | | | | | | | | |
| CLASS | 6 | | | | | | | | | |
| CENTRE NUMBER | S | | | | | INDEX NUMBER | | | | |

## H1 CHEMISTRY — 8872/02

Paper 2

**13 September 2011**

**2 hours**

Candidates answer Section A on the Question Paper.

Additional Materials: Data Booklet

**READ THESE INSTRUCTIONS FIRST**

Write your name, class, Centre number and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

**Section A**
Answer **all** the questions.

**Section B**
Answer **two** questions on separate answer paper.

A Data Booklet is provided. Do not write anything on it.

The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

| For Examiner's Use | |
|---|---|
| Section A | |
| B5 | |
| B6 | |
| B7 | |
| Total | |

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## Section A

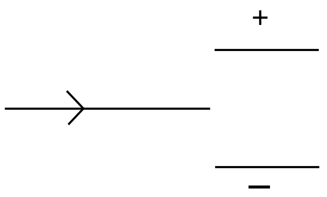
Answer **all** questions in this section.

**1.** **(a)** Complete the table below.

| Element | Electric charge | Mass number | Number of | | |
|---|---|---|---|---|---|
| | | | Protons | Electrons | Neutrons |
| **N** | 0 | | 7 | | 7 |
| **O** | 2- | 18 | | 10 | |
| **F** | | 19 | 9 | 8 | |

**[2]**

**(b)** Beams consisting of the particles listed in the table above are subjected to an electric field as shown in the diagram below. Sketch, and label, on the diagram below to show how beams of each of the three particles are affected by the electric field.




**[3]**

**(c)** **(i)** Write an equation to define the first ionisation energy of nitrogen.

**(ii)** Sketch a graph of the first ionisation energy of the elements of Period 2.

**(iii)** Explain the difference between the first ionisation energy of

I nitrogen and oxygen

II oxygen and fluorine **[5]**

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**2** Chlorine is a pale green gas at room temperature and pressure. There are two stable isotopes of chlorine, $^{35}Cl$ and $^{37}Cl$. In addition, a radioactive isotope, $^{36}Cl$, exists in trace amounts in the environment and can be used for radioactive dating.

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Chlorine is toxic. It can react with water to form a mixture of hydrochloric acid and hypochlorous acid, $HOCl$. As such, it causes irritation to the respiratory system if inhaled as it can dissolve in moist surfaces of the air passages and the lungs.

Chlorine can be detected by smell at 3 ppm or parts per million (1 ppm = 1 mg $dm^{-3}$). At a concentration of 60 ppm, lung damage may occur. Death can result from inhaling chlorine gas at levels above 1000 ppm.

**(a)** A sample of chlorine gas consisting of the isotopes $^{35}Cl$ and $^{37}Cl$ was found to have a relative atomic mass of 35.5. This sample was subsequently contaminated with the radioactive isotope $^{36}Cl$ The relative atomic mass of the chlorine in the contaminated sample was found to be 35.6. Find the percentage abundance of [3]

$^{36}Cl$ present in the contaminated sample.

**(b)** Chlorine is a strong oxidising agent and is often used as a disinfectant in swimming pools. Due to the danger and inconvenience of storing pure chlorine, chlorine that is used for disinfecting swimming is often produced by reacting controlled amounts of hydrochloric acid with sodium hypochlorite.

$$2HCl(aq) + NaOCl(s) \rightarrow Cl_2(g) + H_2O(l) + NaCl(aq)$$

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In an accident, 2 dm$^3$ of 10 mol dm$^{-3}$ $HCl(aq)$ was mixed with 1 kg of solid $NaOCl$. The incident occurred in a room of volume 50 m$^3$. The room was evacuated and equipment was brought in to ventilate the room. The equipment was able to ventilate the building at a rate of 5 m$^3$ min$^{-1}$.

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**(i)** Determine the oxidation number of chlorine in $NaOCl$.

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**(ii)** Calculate the amount of chlorine gas produced.

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**(iii)** Determine if the concentration of chlorine in the room is greater than the lethal concentration of 1000 ppm.

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(iv) Determine the theoretical volume of air needed to dilute the chlorine gas to a concentration lower than 3 ppm.

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(v) The company supplying the ventilation equipment claimed that the equipment would be able to remove the chlorine gas from the room after ten minutes. Suggest if this is a realistic claim.

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3 Many biological processes only occur within a narrow range of pH values. The pH of different fluids found in the body is given below:

| | |
|---|---|
| Saliva | pH 6.8 |
| Blood | pH 7.4 |
| Stomach juices | pH 1.0 to pH 3.0 |
| Intestinal juices | pH 8.5 |

**(a)** Calculate the hydroxide ion concentration in intestinal juices.

[2]

**(b)** The pH of stomach juices is low due to the presence of hydrochloric acid. Hydrochloric acid can be classified as a *strong Bronsted acid*. Explain what you understand by the terms in italics. [2]

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**(c)** The body maintains the pH of blood within a narrow range of values. Death could result if the blood pH decreases below 6.8 or rises above 7.8. The need to maintain the pH within a narrow range of values requires the use of a buffer. In blood, the main buffering system is the $H_2CO_3$ / $HCO_3^-$ buffer.

**(i)** Explain what is meant by the term *buffer*.

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**(ii)** Write equations to show how the $H_2CO_3$ / $HCO_3^-$ buffer functions.

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**(iii)** During heavy exercise, acidic waste products are produced in the body. One function of blood is to remove such waste products. One such response is an increased rate of breathing. In addition to increasing the supply of oxygen to the body, heavy breathing helps to remove a gas which in turn helps to regulate the pH of the blood.

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State the identity of the gas removed and suggest an explanation how heavy breathing helps to regulate blood pH. Include an equation in your answer. [6]

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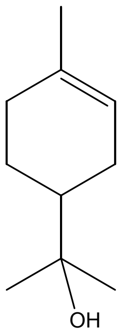
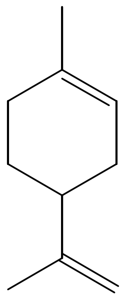
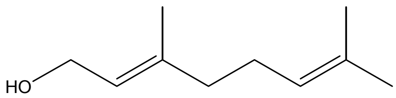
**4** Terpineol, limonene, citral and geraniol are members of a class of compounds called terpenes which are widely found in nature.



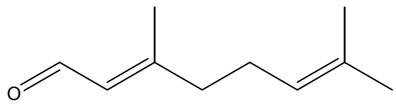
**terpineol**



**limonene**



**geraniol**



**citral**

**(a)** Identify the functional groups present in geraniol and citral. [2]

gerananiol : ……...………………………………………………………………….

citral: ………...………………………………………………………………….

**(b)** State the reagents and conditions by which you could convert geraniol to citral. [1]

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**(c)** Describe, by means of simple chemical tests, how you would distinguish between the following pairs of compounds.

**(i)** geraniol and citral

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**(ii)** terpineol and limonene [4]

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**(c)** Both terpineol and limonene are liquids at room temperature. State and explain if terpineol has a higher boiling point than limonene. [3]

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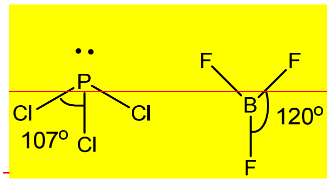
## Section B

Answer **two** of the three questions in this section on separate answer paper.

5~~1.~~ Phosphorus forms three chlorides, $P_2Cl_4$, $PCl_3$ and $PCl_5$, of which the latter two are most common. $PCl_3$ is made industrially on a large scale by the direct chlorination of phos~~f~~phorus, while $PCl_5$ serves as a valuable chemical intermediate in many processes.

[8]

**(a)** **(i)** $PCl_3$ can function as a Lewis base, reacting with boron trifluoride, $BF_3$, in the molar ratio of 1:1. Using the Valence Shell Electron Pair Theory, draw diagrams to illustrate the molecular shapes of $PCl_3$ and $BF_3$ and indicate the bond angles.

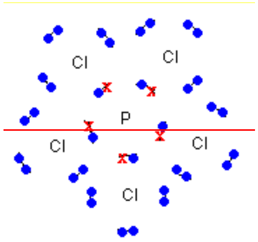



~~[1] for each correct structure~~

**(ii)** Suggest the type of bonding that can take place between $PCl_3$ and $BF_3$.

~~Dative/coordinate bonding. [1]~~

**(iii)** Draw a dot-and-cross diagram of phosphorus pentachloride, $PCl_5$, predicting the shape of the molecule.




**[1]**

~~Shape: Trigonal bipyramidal [1]~~

**(iv)** $PCl_5$ is molecular in the gaseous phase. However, it adopts an ionic structure in the crystalline phase, with the lattice consisting of $[PCl_4]^+$ and $[PCl_y]^-$ ions.

Suggest a likely identity of the ion $[PCl_y]^-$.

~~A likely identity of the ion is $[PCl_6]^-$. [1]~~

**(v)** When dissolved in water, $PCl_5$ gives a strongly acidic solution. Explain why this happens and provide a relevant equation.

~~$PCl_5$ undergoes hydrolysis in water to give strongly acidic solutions (pH 2). P uses its energetically accessible vacant 3d-orbitals for dative bonding with water molecules. [1]~~

~~$PCl_5(s) + 4H_2O(l) \rightarrow H_3PO_4(aq) + 5HCl(aq)$ [1]~~

Elements from Period 3 form oxides that exhibit different physical and chemical properties.

| Compound | $Na_2O$ | MgO | $SiO_2$ | $P_4O_6$ |
|---|---|---|---|---|
| Melting point / °C | 1132 | 2800 | 1700 | 24 |

[4]

**(b)** **(i)** Explain why MgO has a higher melting point than $Na_2O$ in terms of structure and bonding.

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~~Both MgO and $Na_2O$ have **giant ionic structures[✓]** and **strong electrostatic forces of attraction exist between their ions.[✓]** Due to the **higher charge density of $Mg^{2+}$**,**[✓] more energy is needed[✓]** to break the stronger electrostatic attraction between $Mg^{2+}$ and $O^{2-}$ ions, resulting in a higher melting point than $Na_2O$.~~

~~**4[✓]: 2 mark**~~
~~**2-3[✓]: 1 mark**~~

**(ii)** Explain why $P_4O_6$ has a lower melting point than $SiO_2$ in terms of structure and bonding.

~~$P_4O_6$ has a **simple covalent structure[✓]** and weak **van der Waals forces exist between its molecules.[✓]** On the other hand, $SiO_2$ has a **giant covalent structure[✓]** and strong **covalent bonds exist between its atoms.[✓]** Thus, **less energy[✓]** is needed to break the van der Waals forces between $P_4O_6$ molecules, resulting in a lower melting point.~~

~~**5[✓]: 2 marks**~~
~~**3-4[✓]: 1 mark**~~

**(c)** **(i)** Define *standard enthalpy change of formation.* ~~[8]~~

~~The standard enthalpy change of formation of a substance is the **enthalpy change** when **one mole** of the substance is **formed from its constituent elements in their standard states** (under standard conditions). **[1]**~~

**(ii)** Using relevant data from the *Data Booklet*, complete the energy cycle by stating **P** to **R**. Include state symbols or units in your answers when necessary.

Therefore, calculate the enthalpy change of formation of magnesium oxide.

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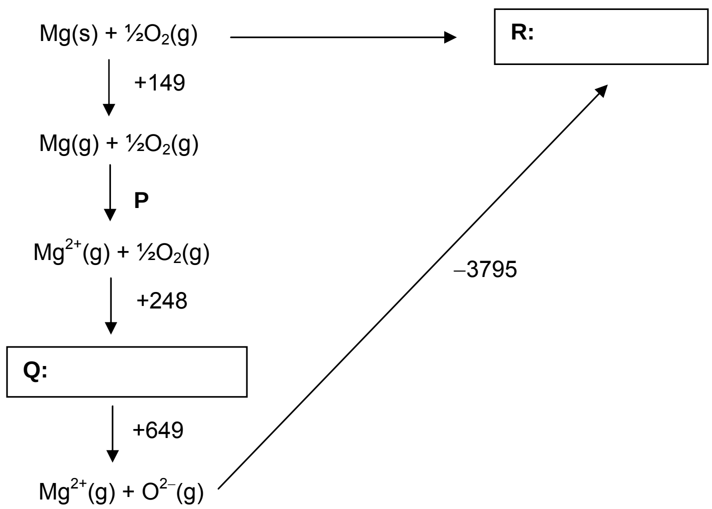


~~Enthalpy change of formation of MgO = +149 + 2186 + 248 + 649 – 3795~~ [5]
~~= **–563 kJ mol$^{-1}$** **[1]**~~

Liquid butane is commonly found in canisters used for portable gs stoves during camping trips.

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**(d)** When 1 g of butane was burnt under a container carrying 150 g of water, it was found that the temperature of water rose from 25 °C to an unknown temperature, **T**. Given that the enthalpy change of combustion of butane is –2877 kJ mol$^{-1}$ and the burning process is 80 % efficient, find the unknown temperature **T**. (Specific heat capacity of water = 4.2 J g$^{-1}$ K$^{-1}$) [3]

~~Amount of limiting reagent (butane) = 1 / 58~~
~~= 0.01724 mol **[1]**~~

~~When burning process is 100 % efficient,~~
~~Heat evolved = 150 x 4.2 x (T – 25) x 100/80~~
~~= 787.5 (T – 25) J **[1]**~~

~~$\Delta H_c$(butane) = Heat evolved / amount of butane~~

~~–2877 x 10$^3$ = –787.5 (T – 25) / 0.01724~~
~~T = 88.0 °C [1]~~

**[Total: 20]**

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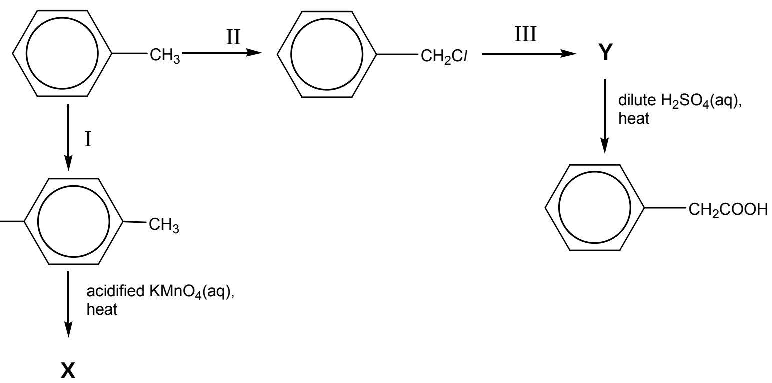
~~6~~2~~.~~ **(a)** A reaction scheme starting from methylbenzene is shown below. Draw the structures of the organic products ~~A~~, X and Y~~to E~~ and state the reagents and conditions needed for steps **I** to ~~V~~**III**. [5]




~~Step **I**: $Cl_2$, Fe/$FeCl_3$/$AlCl_3$ catalyst [✓]~~
~~Step **II**: $Cl_2$, uv light [✓]~~
~~Step **III**: ethanolic KCN, heat [✓]~~
~~Step **IV**: $CH_3CH_2OH$, a few drops of conc $H_2SO_4$, heat under reflux [✓]~~
~~Step **V**: $LiAlH_4$ in dry ether [✓]~~

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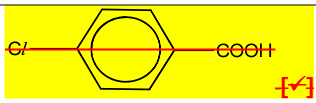
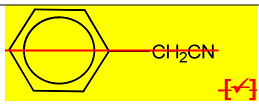
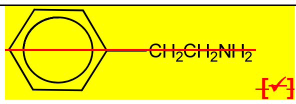
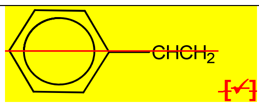
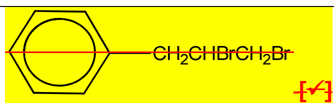
| ~~A:~~ |  |
| ~~B:~~ |  |
| ~~C:~~ |  |
| ~~D:~~ |  |
| ~~E:~~ |  |

~~**2[✓]: 1 mark**~~

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**(b)** Rank the following compounds in order of increasing acidity. Explain your answer.

Propanoic acid, propanol, 2-chloropropanol, 2-bromopropanoic acid [4]

~~Acidity: **Propanol < 2-chloropropanol < propanoic acid < 2-bromopropanoic acid [1]**~~

~~Acid strength depends on stability of anion. The more stable the anion, the stronger the acid.~~

~~In 2-chloropropanol, the **electron-withdrawing *Cl* atom disperses the negative charge on the alkoxide anion,[✓]** stabilising the anion. On the other hand, in propanol, the **electron-donating alkyl group intensifies the negative change on the alkoxide anion,[✓]** destabilising the anion.~~

~~In propanoic acid, the **negative charge on the carboxylate anion is delocalised into the COO$^-$ group,[✓] stabilising the anion to a larger extent as compared to the alkoxide anion.[✓]** In 2-bromopropanoic acid, the **negative charge on the carboxylate anion is both delocalised into the COO$^-$ group and dispersed by the electron-withdrawing Br atom,[✓]** thus stabilising the anion to the largest extent.~~

~~**5[✓]: 3 marks**~~
~~**4[✓]: 2 marks**~~
~~**3[✓]: 1 mark**~~

**(c)** The following equation represents ~~the rate of~~ the reaction between 2-bromo-2-methylpropane and aqueous sodium hydroxide when heated under reflux.

$$(CH_3)_3CBr + OH^- \rightarrow (CH_3)_3COH + Br^-$$

| Experiment | Initial $[(CH_3)_3CBr]$ / mol dm$^{-3}$ | Initial $[OH^-]$ / mol dm$^{-3}$ | Initial rate of reaction / mol dm$^{-3}$ s$^{-1}$ |
|---|---|---|---|
| 1 | $1.0 \times 10^{-3}$ | $1.0 \times 10^{-1}$ | $0.5 \times 10^{-3}$ |
| 2 | $2.0 \times 10^{-3}$ | $1.0 \times 10^{-1}$ | $1.0 \times 10^{-3}$ |
| 3 | $2.0 \times 10^{-3}$ | $0.5 \times 10^{-1}$ | $1.0 \times 10^{-3}$ |

~~[11]~~

**(i)** Determine the order of reaction with respect to the two reactants. Hence, write the rate equation.

~~Comparing experiments 1 and 2~~
~~Initial rate of reaction is doubled when initial $[(CH_3)_3CBr]$ is doubled~~
~~Order of reaction with respect to $(CH_3)_3CBr$ is 1 **[1]**~~

~~Comparing experiments 1 and 3~~
~~Initial rate of reaction stays the same when initial $[OH^-]$ is halved~~
~~Order of reaction with respect to $OH^-$ is 0 **[1]**~~

~~Rate = k$[(CH_3)_3CBr]$ **[1]**~~

**(ii)** Calculate the value of the rate constant, stating its units clearly.

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~~Using experiment 1~~
~~$0.5 \times 10^{-3} = k \times 1.0 \times 10^{-3}$~~
~~$k = 0.500\ s^{-1}$ **[1]**~~

**(iii)** Calculate the initial rate of reaction when the initial concentration of $(CH_3)_3CBr$ is 2.50 x ~~×~~ $10^{-3}$ mol $dm^{-3}$ and the initial concentration of $OH^-$ is 2.0 x ~~×~~ $10^{-1}$ mol $dm^{-3}$.

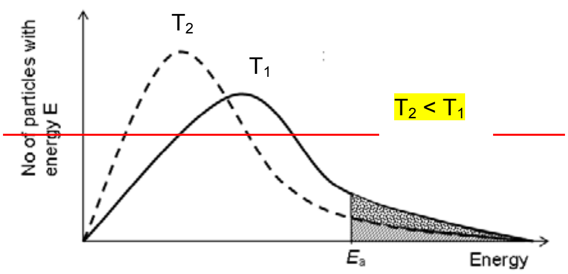
~~Rate = $0.500 \times 2.50 \times 10^{-3}$~~
~~= $1.25 \times 10^{-3}$ mol $dm^{-3}\ s^{-1}$ **[1]**~~

**(iv)** How would you expect the initial rate of reaction to change when 2-bromo-2-methylpropane is replaced by 2-chloro-2-methylpropane? Explain your answer.

~~The initial rate of reaction will be **lower,[1]** due to the **breakage of the stronger C–*Cl* bond.[1]**~~

**(v)** With the aid of a Maxwell-Boltzmann distribution curve, explain what happens to the rate of reaction if the reaction mixture is surrounded by ice-cold water?

[11]




~~Correct shape of graph [✓]~~
~~Correct labeling of axes [✓]~~

~~At a **lower temperature,[✓]** there is a **smaller number of reactant particles with energy greater than or equal to the activation energy,[✓]** resulting in a **decrease in the frequency of effective collisions[✓]** between the reactant particles and hence a **lower rate of reaction.[✓]**~~

~~**6[✓]: 4 marks**~~
~~**5[✓]: 3 marks**~~
~~**3-4[✓]: 2 marks**~~
~~**1-2[✓]: 1 mark**~~

**[Total: 20]**

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7 Esters are derivatives of carboxylic acids and are widely found in nature. Esters are also widely used in industry. [10]

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**(a)** Ethyl ethanoate is present in wine and is also widely used as a solvent. Ethyl ethanoate can be synthesised from ethanol and ethanoic acid in the presence of concentrated sulfuric acid.

$$CH_3CH_2\,OH + CH_3COOH \xrightleftharpoons{\text{conc } H_2SO_4} CH_3COOCH_2CH_3 + H_2O$$

**(i)** Write an expression for $K_c$ for the above reaction and state its units.

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**(ii)** A student carried out an experiment to determine $K_c$ for the above reaction. 12.00 g of glacial ethanoic acid, 9.20 g of ethanol were reacted together in the presence of concentrated sulfuric acid. At equilibrium, the yield of ethyl ethanoate was 33.5%. ~~and the total volume of the reaction mixture was 44.0 cm~~$\cancel{^3}$~~.~~

Calculate the value for $K_c$.

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**(iii)** $K_c$ for the above reaction increases with increasing temperature. State and explain if forward reaction is endothermic or exothermic.

**(iv)** A student repeated the above experiment but used an ethanol-water mixture instead of pure ethanol. Explain how the yield of the reaction will be affected.

**(v)** The concentrated sulfuric used in the reaction has two roles. State these two roles. [10]

**(b)** When an ester, **A**, $C_{10}H_{18}O_4$, is heated under reflux with dilute aqueous acid, **B**, $C_2H_2O_4$, and **C**, $C_4H_{10}O$, are produced.

Both **B** and **C** react with sodium metal to produce a colourless gas.

When **B** is heated under reflux with acidified potassium manganate(VII), a colourless gas is evolved. No organic product is produced.

**C** reacts with acidified potassium manganate(VII) to produce **D**.

Both **C** and **D** react separately with aqueous alkaline iodine to produce a yellow [10] precipitate.

**D** reacts with 2,4-dinitrophenylhydrazine to produce, **E**.

Deduce the structures of **A, B, C, D** and **E** and explain the chemistry of the reactions involved.

**[Total: 20]**

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